

The Utility Problem

In electrical utilities, meter-to-transformer association data refers to the relationship between a specific meter (used to measure electricity consumption) and the transformer that serves it. This association data is important for utilities due to multiple reasons:

- Distribution or Asset management - this data is used to monitor and manage the distribution system by providing information on the load on each transformer and identifying potential issues such as overloaded transformers or power outages.
- Planning and forecasting - this data can also be used to plan and forecast future energy needs and to optimize the distribution system by identifying areas where additional capacity is needed (transformer upgrades or replacements).
- Customer Service - with accurate association data, utilities can promptly identify and address issues reported by customers, such as power outages or service disruptions.

Overall, meter-to-transformer association data is an important part of managing and operating electrical utilities, as it allows for efficient management of the distribution system and improved customer service. As the meter-to-transformer association data plays a major role in utilities, so the accuracy of this data is important as well. Unfortunately, inaccurate meter-to-transformer association data is nothing unusual for utilities nowadays. Numerous errors with different levels of occurrence exist from utility to utility.

For these reasons, it is important to have a process in place which will allow for easy, fast and inexpensive meter-to-transformer association data verification and correction.

The Data Solutions and Methods

There are multiple methods how for verifying and consequently correcting the meter-to-transformer association data. Each method bears its own characteristics in terms of cost and resulting accuracy. Here are some examples:

- On-site inspections - very accurate but costly.
- GIS analysis techniques - for example, elevation analysis, distance analysis, etc.
- Advanced Metering Infrastructure (AMI) - advanced analysis of the voltages and currents from AMI.
- Transformer Loading Analysis - specific loading of the parent transformer might imply the errors in meter-to-transformer association data.

Furthermore, it is fair to say that each method alone will not be the best for the needs of each utility. It is most likely the combination of mentioned methods which, when combined, will yield optimal results in terms of accuracy, complexity and cost. It is also important to note that the Awesense Energy Transition Platform offers powerful methods of data access to address the meter-to-transformer association data verification and correction. The following description will shortly describe each of the data methods and how the Awesense Platform can be helpful.

GIS Analysis Techniques:

The Awesense Platform's geospatial component is very helpful since all the grid elements in the Awesense Platform contain geospatial information. Both analyses assume a higher degree of meter coordinate accuracy. With this, two basic analyses can be performed:

1. Elevation analysis - for the geographies where the streets are on steeper geographies, it is beneficial to analyze the elevation of the meter and parent transformer. For situations with higher than acceptable differences, the flag can be raised, indicating potential association error.
2. Distance analysis - secondary distribution (low voltage) lines have some distance range where the child meter can be located compared to the parent transformer. All occurrences where the distance is too big (and outlier) can be flagged as association errors.

For both analyses, Awesense's Energy Data Model (EDM) SQL API offers convenient PostGIS functions (ST_Closest...) for finding the closest geometrical elements.

Advanced Metering Infrastructure (AMI):

Advanced analysis of the AMI voltage registers (average, min, max voltage registers) can be put into practice to identify the wrong association and also suggest a correction. More about this method is described in the Awesense use case "Phase Identification Analysis".

Transformer Loading Analysis:

Transformer loading quantifies the loading of the transformer over time compared to its rated power (kVA). Some heavily overloaded and/or significantly underloaded transformers might indicate errors in meter-to-transformer association data. More about this analysis can be seen in the Awesense use case "Transformer Loading Analysis".

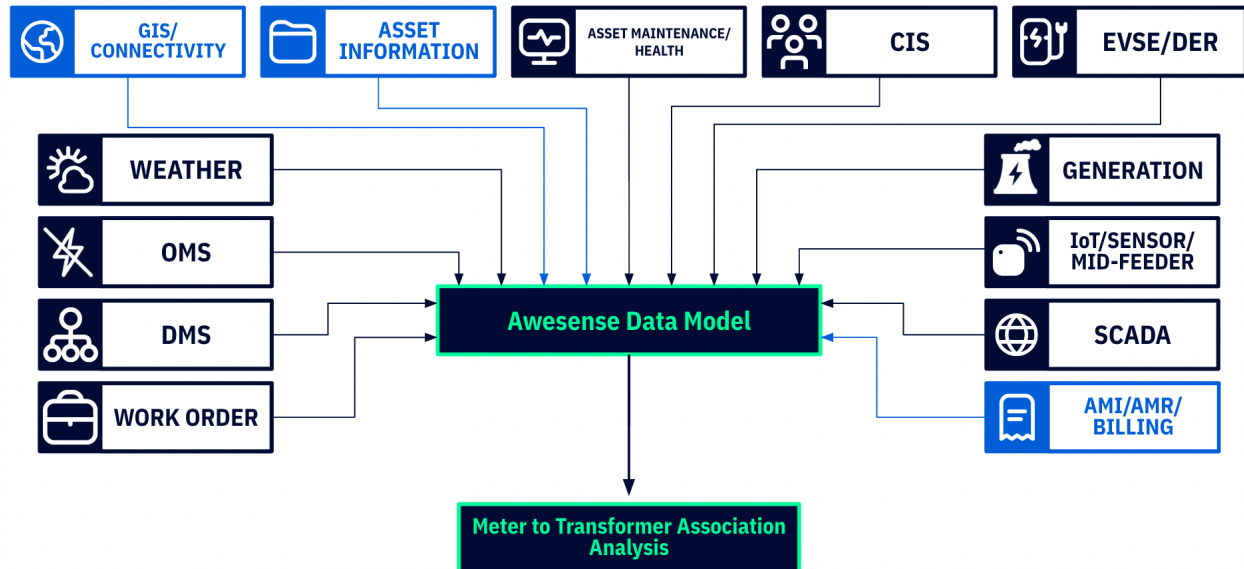
Meter to Transformer Association Analysis

[UC42-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.